Omid Tarabavar

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Research Interests

- **Multimodal Machine Learning**: Multimodal representation learning and fusion of textual, visual, and contextual information, particularly for fake news detection and fact verification.
- **Social Computing**: Misinformation and information credibility in online communities, and how social context and user interactions shape the spread and detection of misleading content.
- **Large Language Models and Agentic AI**: Evaluation and benchmarking of LLMs, and the design and optimization of agentic systems, including planning, memory, and task routing.
- **Trustworthy Machine Learning**: Model reliability and robustness under distribution shift, with a focus on generalization to unseen data and environments.

Education

Amirkabir University of Technology (Tehran Polytechnic) — Tehran, Iran

M.Sc. in Artificial Intelligence & Robotics, **GPA: 18.86/20 | Current Class Rank: 1st** — Oct 2025 – Present

Thesis: Fake News Detection Using Multimodal RAG and Learned Query Representation

Advisors: Dr. Saeedeh Momtazi and Dr. Reza Safabakhsh

Developing a multimodal RAG framework that learns query representations for evidence retrieval and combines textual and visual information to assess news credibility.

Selected Coursework: Natural Language Processing (20/20), Machine Learning (20/20), Trustworthy Machine Learning (18.33/20), Deep Learning (18.2/20), Advanced Natural Language Processing (17.7/20).

Shiraz University of Technology — Shiraz, Iran

B.Sc. in Computer Engineering and Information Technology — Sep 2021 – Sep 2025

GPA: 19.09/20 (4.00/4.00) | Graduated 1st in Class

Thesis: Typing Assistant [Code] [Report] [Transcript]

Built a next-word prediction system with probabilistic (n-gram), sequence (LSTM), and attention-based (Transformer) models.

Selected Coursework: Artificial Intelligence (20/20), Fundamentals of Computational Intelligence (20/20), Linear Algebra (20/20), Algorithm Design (19.25/20), Engineering Probability and Statistics (19/20).

Publications

Visual-Anchored Context Fusion for Source-Faithful Evidence Retrieval in Multimodal Fact Verification. [Code]
Omid Tarabavar, Saeedeh Momtazi, and Reza Safabakhsh. Under review at *Applied Soft Computing*.
Proposes VACF, a training-free multimodal evidence retrieval framework that combines protected visual-anchor selection, contextual completion, and source-faithful restoration to preserve evidence provenance.

Selected M.Sc. Course Projects

- **Persian Banking Assistant: Hybrid Intent Routing, RAG, and LLM-as-a-Judge** [Code] — Advanced NLP, 2026
  - Built a multi-turn Persian banking chatbot in which a fine-tuned ParsBERT intent classifier routes low-confidence queries to a few-shot LLM fallback (accuracy 87.7% → 96.2%), with LLM-based slot filling and hybrid dense + BM25 retrieval with cross-encoder reranking.
  - Evaluated the system with RAGAS and a custom Gemma-based LLM-as-a-judge, showing that the two agree on answer correctness ($r = 0.74$) but barely on faithfulness ($r = 0.19$); tested improvements with paired significance tests.
- **Parameter-Efficient Fine-Tuning and Catastrophic Forgetting in Gemma 3n** [Code] — Advanced NLP, 2026
  - Fine-tuned Gemma 3n on Persian news with LoRA and with bottleneck adapters implemented from scratch in PyTorch. The adapters used 64% fewer trainable parameters, reached lower in-domain perplexity (5.97 vs. 6.06), and forgot less out of domain (+19.9% vs. +33.6% Wikipedia perplexity).
- **Adversarial Robustness and Fairness Audit** [Code] — Trustworthy ML, 2026
  - Implemented FGSM/PGD attacks and adversarial training on Fashion-MNIST and CIFAR-10, and ruled out gradient masking with strong PGD, EOT-PGD, transfer attacks, and AutoAttack (CIFAR-10 AutoAttack accuracy 0% → 30.7%).
  - Audited income classifiers on Adult for group and intersectional fairness, calibration, SHAP explanations, and counterfactual recourse, and compared the trade-offs of reweighing and threshold-based mitigation.
- **Generalization Under Distribution Shift** [Code] — Trustworthy ML, 2026
  - Compared importance reweighting, DANN, and CycleGAN for unsupervised domain adaptation on rotated MNIST, showing that all fail under severe (90°) shift, and studied layer-wise transfer from 200 labeled target samples (80.2% from scratch → 96.1%).

Teaching Experience

- **Teaching Assistant**, Sharif University of Technology
  - Natural Language Processing (Fall 2026, *graduate*) — Dr. Mahdi Samiei
- **Teaching Assistant**, Amirkabir University of Technology
  - Deep Learning (Fall 2026, *graduate*) — Dr. Reza Safabakhsh
- **Teaching Assistant**, Shiraz University of Technology
  - Algorithm Design (Spring 2024, *undergraduate*) — Dr. Omid Bushehrian
  - Formal Languages and Automata (Spring 2024, *undergraduate*) — Dr. Ali Sadeghi
  - Data Structures (Fall 2023, *undergraduate*) — Dr. Rasool Esmaeilyfard
  - Advanced Programming (Spring 2023, *undergraduate*) — Dr. Pirooz Shamsinejad
- **Instructor, ICPC Preparation**, Shiraz University of Technology — Fall 2024
  - Taught the fundamentals of competitive programming to prepare students for the 25th Asia Regional ICPC Contest in Iran.

Work Experience

- **Machine Learning Developer**, Modern Cybernetics — Shiraz, Iran
  On-site — Jan 2025 – Jun 2025
  - Developed AI agents using fine-tuned large language models and retrieval-augmented generation (RAG), with data collected from web sources by automated scraping tools.
- **Deputy Secretary**, AI Scientific Association, Shiraz University of Technology — Shiraz, Iran
  Student association — May 2023 – Jul 2024
  - Organized AI events, workshops, and competitions for students, including preparation for programming contests such as ICPC.

Honors and Awards

- **Ranked 1st** in the M.Sc. class at Amirkabir University of Technology (current) and **graduated 1st** in the B.Sc. class at Shiraz University of Technology.
- **Direct admission** to the M.Sc. program in Artificial Intelligence at Amirkabir University of Technology based on academic excellence.
- **Ranked within the top 2%** among 126,000+ participants in the National University Entrance Exam.
- **Admitted to NODET High Schools** (National Organization for Development of Exceptional Talents) through a competitive entry exam.

Online Courses

- **Linear Algebra (18.06)** [Notes] – *MIT, Gilbert Strang* — Jul 2025 – Sep 2025
- **Deep Learning Specialization** – *DeepLearning.AI, Andrew Ng et al.* — Feb 2025 – Mar 2025
- **Natural Language Processing Specialization** – *DeepLearning.AI, Łukasz Kaiser et al.* — Sep 2024 – Oct 2024
- **Machine Learning Specialization** [Certificate] – *DeepLearning.AI, Andrew Ng et al.* — Sep 2024 – Oct 2024
- **Mathematics for Machine Learning and Data Science Specialization** – *DeepLearning.AI* — Jul 2024 – Aug 2024

Skills

- **Large Language Models**: Transformer architectures (encoder, decoder, encoder–decoder), pretraining and fine-tuning, parameter-efficient fine-tuning (LoRA, adapters), instruction tuning and alignment (RLHF, DPO), prompting, in-context learning and chain-of-thought, retrieval-augmented generation, LLM agents and tool use, LLM evaluation (human, benchmark-based, LLM-as-a-judge), quantization and distillation
- **Natural Language Processing**: Language modeling (n-gram to neural), word embeddings, text classification, sequence labeling (POS tagging, NER), syntactic parsing, machine translation, question answering
- **Multimodal and Trustworthy ML**: Vision-language models, multimodal retrieval and fusion, adversarial robustness, fairness and explainability, domain adaptation
- **Programming Languages**: Python, Java, C++, C
- **Frameworks and Tools**: PyTorch, Hugging Face ecosystem, TensorFlow/Keras, Scikit-Learn, Pandas, NumPy, NLTK, Matplotlib, Git, Linux, LaTeX, MySQL
- **English**: IELTS practice test (Aug 2024): Overall 7.0 (unofficial)

References

- **Dr. Saeedeh Momtazi** – Associate Professor, Amirkabir University of Technology — momtazi@aut.ac.ir
- **Dr. Reza Safabakhsh** – Professor, Amirkabir University of Technology — safa@aut.ac.ir
- **Dr. Mahdi Samiei** – Sharif University of Technology — mm.samiei@sharif.edu
- **Dr. Omid Bushehrian** – Associate Professor, Shiraz University of Technology — bushehrian@sutech.ac.ir
- **Dr. Ali Sadeghi** – Assistant Professor, Shiraz University of Technology — sadeghi@sutech.ac.ir
- **Dr. Rasool Esmaeilyfard** – Assistant Professor, Shiraz University of Technology — esmaeily@sutech.ac.ir
- **Dr. Pirooz Shamsinejad** – Assistant Professor, Shiraz University of Technology — p.shamsinejad@sutech.ac.ir